RANGE DESK USER MANUAL

Ballistics, truing and match planning for F-Class at Bisley. Version of 4 November 2026.

The app is at **https://dcownley.github.io/range-desk/**. It runs in Chrome on a tablet, phone or computer, and works without signal once it has been opened at the range. It has two views:

- **Full view**, for the desk: the map, match planning, the load, imports and analysis.
- **Range mode** (the red button), for the firing point: the essentials in large type, shown in the order you need them.

> **The principle.** You choose the rifle, load, range, target and time. The app works out the firing point, direction of fire, Coriolis, forecast and sun. On the day it measures and trues: velocity from the Garmin, wind from the Calypso or Kestrel, and elevation and wind from where your shots land. Range functionality comes first; every input is one tap away but never in the way.

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1 GETTING STARTED AND UPDATING

- Open the address in **Chrome**. On a tablet, add it to the home screen for one-tap access.
- **Updating:** on GitHub (dcownley/range-desk), go to **Add file → Upload files**, choose **index.html**, then **Commit**. GitHub can take up to 10 minutes to serve the new file. To fetch it straight away, open the address with `?v=` and any new number on the end, for example `…/range-desk/?v=5`. A new version shows its new buttons.
- **Before a range day:** open the app once with signal at the range, or tap **Save this range for offline**, so the app, weather and map are kept on the device.
- **Help** (header in the full view, Menu in Range mode) explains every card.

2 YOUR LOAD AND RIFLES

Load (full view → Load button in the header, or the Load card; in Range mode, Menu → Load)

The Load card sits under the Match plan. Choose the load from the list, edit it under **Edit this load**, or tap **New load** (or **Duplicate** to copy the current one and change only what differs). Changes save as you type.

Input	Notes
Name, calibre, bullet, weight	For example "KW Precision 7/6.5 PRC G238", Berger 180 gr
Drag model and BC	G7 0.349 (load), or as trued from your records
Muzzle velocity and its temperature	2900 fps at 15 °C; temperature sensitivity in fps per °C once learned
Zero, sight height, twist and direction	109 yd zero; 1:8.5 right-hand
Scope unit and step	MOA, ¼ or ⅛. Every value is shown in MOA as on the turret.

Rifles (full view → Match plan → Rifles)

Add each rifle by **make and model** and **serial number**. Each planned detail records the rifle and load used, and the session tags carry the rifle into every string.

Ammunition batch (Session details)

Enter the batch or lot, and change it whenever you change ammunition. Velocity, truing and temperature learning never mix batches, and Records shows a By batch table.

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3 RANGES: FIRING POINTS AND TARGETS

Targets always run from **1 on the left** to the highest number on the right, looking downrange. The **direction of fire** is from the firing point to the target, and every target on a range shares it. It drives Coriolis, the crosswind and headwind from the forecast, and the sun's position. Once a range is set up, entering a target number places that target and its firing point automatically.

Range	Targets	Distances	Set-up
Century	1–128 (10 butts)	300–600 yd	Done: from Saturday's target 49 and its 600 yd firing point
Stickledown	1–51 (51 is the Buffalo)	800–1200 yd (1200: lanes 21–33 and 39–50)	Once, at the range, by GPS (below)

Set up a range by GPS at the firing point (recommended)

1. Full view → **Match plan** → open **Ranges**.
2. Choose the **range**, the **distance** of the mound you are on, and the **target number** of the firing point you are standing at.
3. Stand on that firing point and tap **Pin this firing point (GPS)**. Allow location access when asked. The message confirms it was recorded and how accurate the reading is (usually ±3 to 5 m).
4. Walk **30 or more targets along the same mound**, enter that target's number, and tap **Pin this firing point (GPS)** again.
5. The Ranges line shows **"Stickledown ready (two GPS firing points)"**. Every target and firing point on that range is now known.

The same button is in Range mode on the Next detail card, where it uses the loaded detail's target and distance.

Set up a range from the FP and T pins (most accurate on the map)

1. In the header, tap **Place firing point** and tap the firing point on its mound. Then tap **Place target** and tap that lane's target frame.
2. In Match plan → Ranges, choose the range, the **distance** of that mound and the **target number**, then tap **Use the FP and T pins**.
3. The direction of fire is taken exactly from the firing point to the target. A second known target, pinned or marked the same way, sets the spacing between targets; otherwise a standard spacing is used.

The app checks that the two pins are the stated distance apart. Small differences are trued by the rangefinder on the day.

Set up a range on the map (alternative)

- **Pin it on the map:** enter the target number, tap the button, then tap that target's frame in the butts on the satellite map.
- **Use the T pin:** place the T pin on a target frame (Place target), enter its number, and tap **Use the T pin**.
- Pin two targets well apart. Pinning the same number again replaces it.

Check my position

In Range mode, **Check my position (GPS)** names the range, distance and target you are standing at, and how far you are from the planned firing point.

Trued distances, saved automatically

When you range the frame with the Sig on a planned detail, the distance is saved when the string ends. Later plans use it for that target, and the average offset for other targets at the same firing point.

4 MATCH PLAN: SQUADDING AND FORECASTS

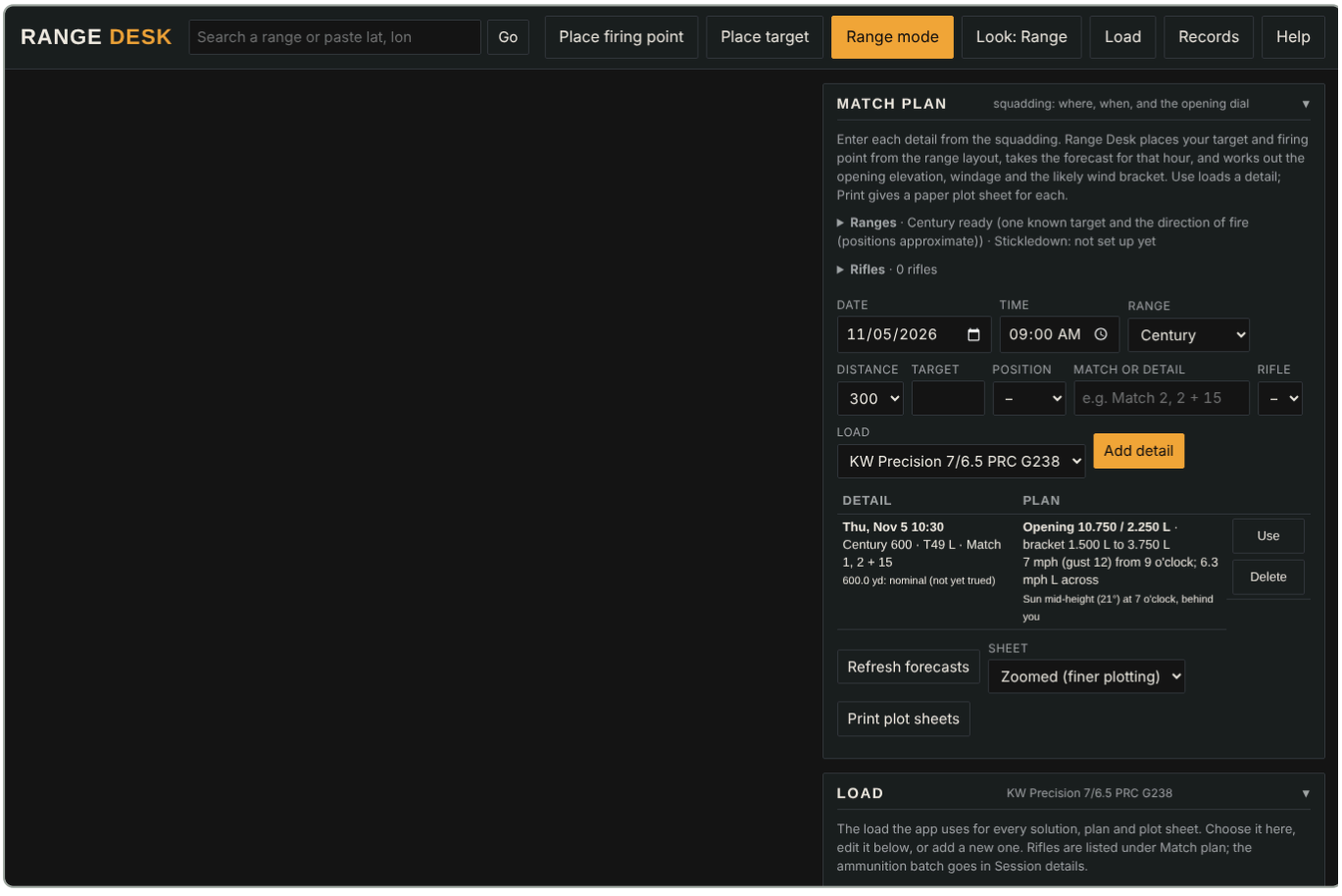
Full view → **Match plan**. Enter each detail from the squadding:

Input	Example
Date and time of the detail	Sat 7 Nov, 09:30
Range and distance	Stickledown, 900
Target number and position	23, left
Match or detail	Match 2, 2 + 15
Rifle and load	From your lists

Tap **Add detail**. For each detail the app shows:

- **Forecast wind** for that hour at the firing point: speed, gusts, clock direction and the crosswind on your line. Your wind truing factor from target plots is applied where you have one.
- **Opening elevation and windage** for that rifle and load in that hour's air.
- **Bracket:** the windage at the lull and at the gust.
- **Sun:** its height and clock position relative to your line, for example "low at 2 o'clock, ahead: glare likely".
- **Distance:** nominal, or trued from an earlier ranging.

Use these buttons on each row: **Use** loads the detail into the app; **Delete**, at the right-hand end of the row (scroll the table sideways), removes it; **Refresh forecasts** fetches the latest forecast. Forecasts reach up to 4 days ahead.



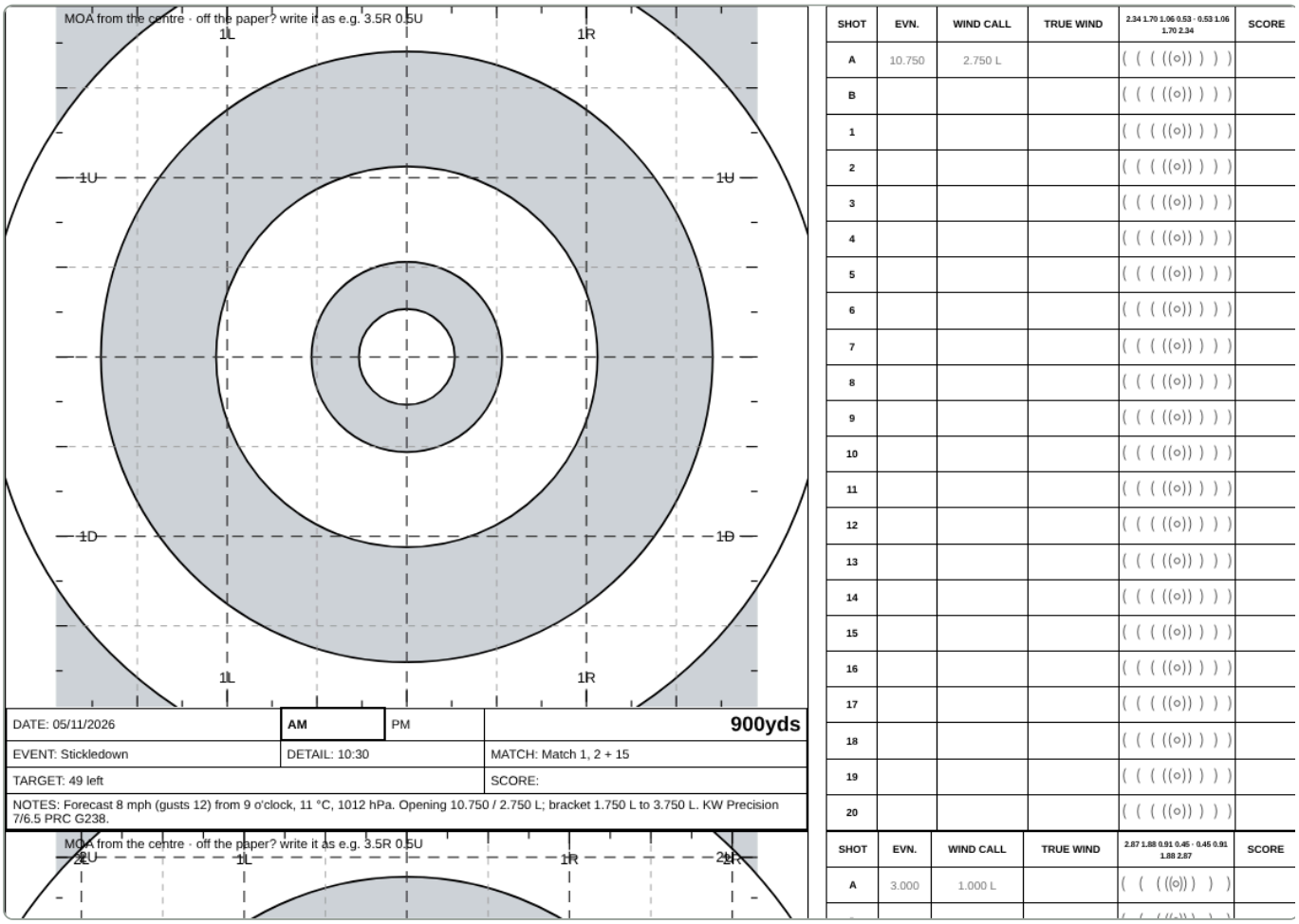
Match plan: details with forecast, opening dial and bracket.

5 PRINTED PLOT SHEETS

Choose **Sheet** (Zoomed for finer plotting, Whole face, or Both), then tap **Print plot sheets**. You get one landscape A4 page per detail, laid out like your own sheets:

- the face to scale, with an MOA grid and MOA scales round the edge (1R, 1L, 1U, 1D). Write a shot that goes off the paper as, for example, "3.5R 0.5U";
- rows A, B and 1–20, with EVN, Wind call, True wind, the wind scale marked at the ring edges in MOA, and Score;
- date, distance, event, detail, match and target filled in, with the forecast, opening dial and bracket in Notes. Row A shows the opening dial lightly.

The 300, 500 and 600 yd details use the 600 yd sheet's scale, as the faces are the same in angle.



A printed zoomed sheet with MOA edge scales.

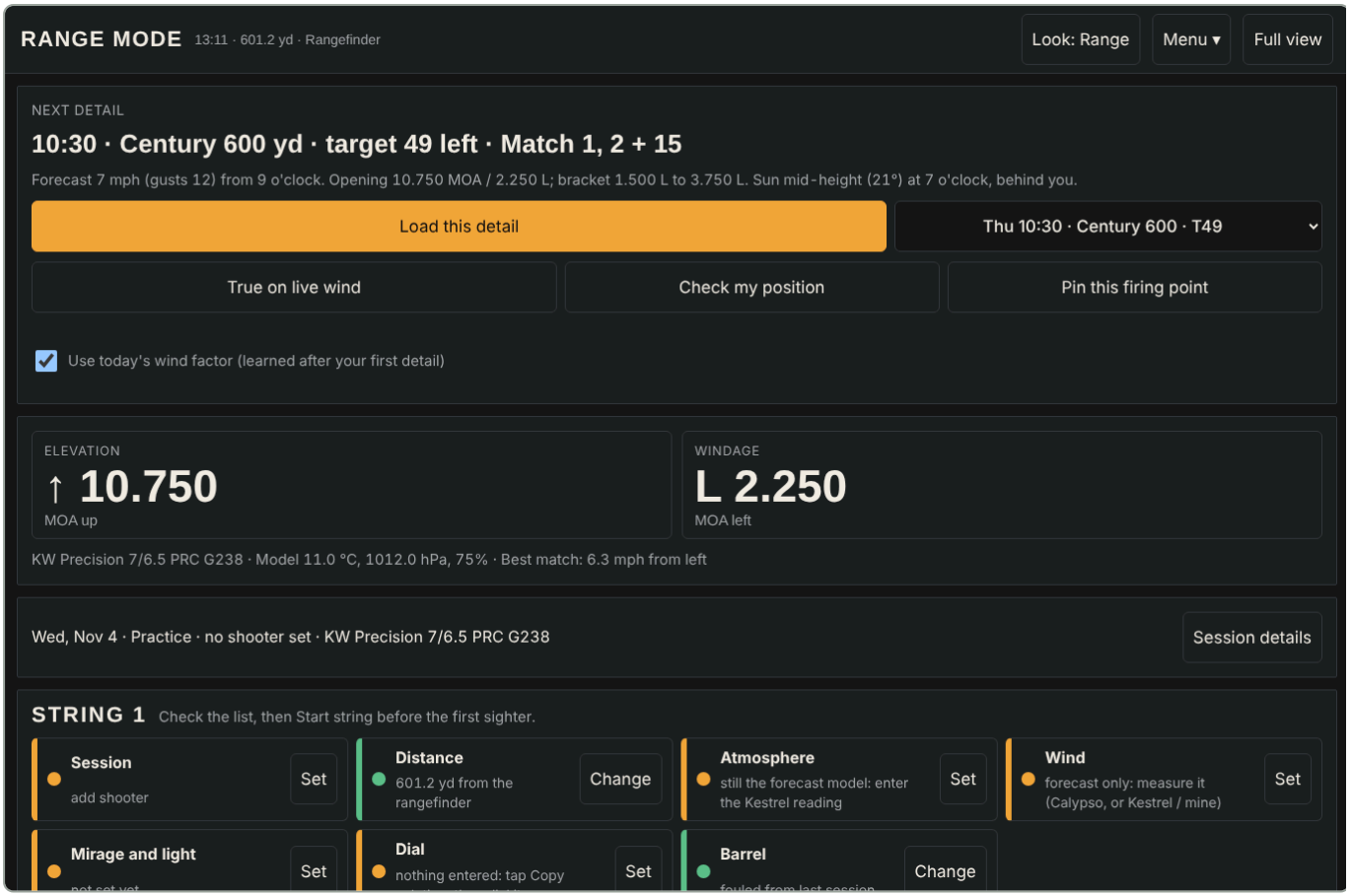
6 RANGE MODE: ARRIVING

- **Look** (header): cycles through Range (graphite, amber), Range green, Sun (paper and ink, for bright light) and Dusk (red on black, to keep night vision).
- **Menu** (header): Load, Records and Help. **Full view** returns to the desk view.
- **Session details:** shooter (name, competitor number, rifle), session type (Practice, Training, Coached, Competition), event, match, coach and **batch**.
- **Today's velocity:** enter the Garmin average, or **Use the last Garmin average**. It drives today's solutions without changing the load, and teaches the app the load's temperature sensitivity over time.
- **Instruments:** connect the Calypso and Weatherdot (section 9). Compare them with the Kestrel.

7 RANGE MODE: SHOOTING A DETAIL

Before the string

1. **Next detail** (top card) shows the next detail from your plan, with its forecast, opening dial, bracket and sun. Tap **Load this detail**, or choose another from the list.
2. **True the forecast on live wind:** compares the Calypso or Kestrel with the forecast for this hour and carries the difference into the coming details, fading over 4 hours. The truing factor from your target plots for that source is then applied.
3. **Checklist:** Session, Distance, Atmosphere, Wind, Mirage and light, Dial and Barrel. Each line shows green, amber or red, and **Set** or **Change** opens just that card. **All set-up cards** shows them all.
4. **Distance:** range the frame with the Sig and enter the line-of-sight figure, not the angle-modified one.
5. **Start string** before the first sighter. Choose sighters (0–3), the number to count (10, 15 or 20), and whether to convert sighters.

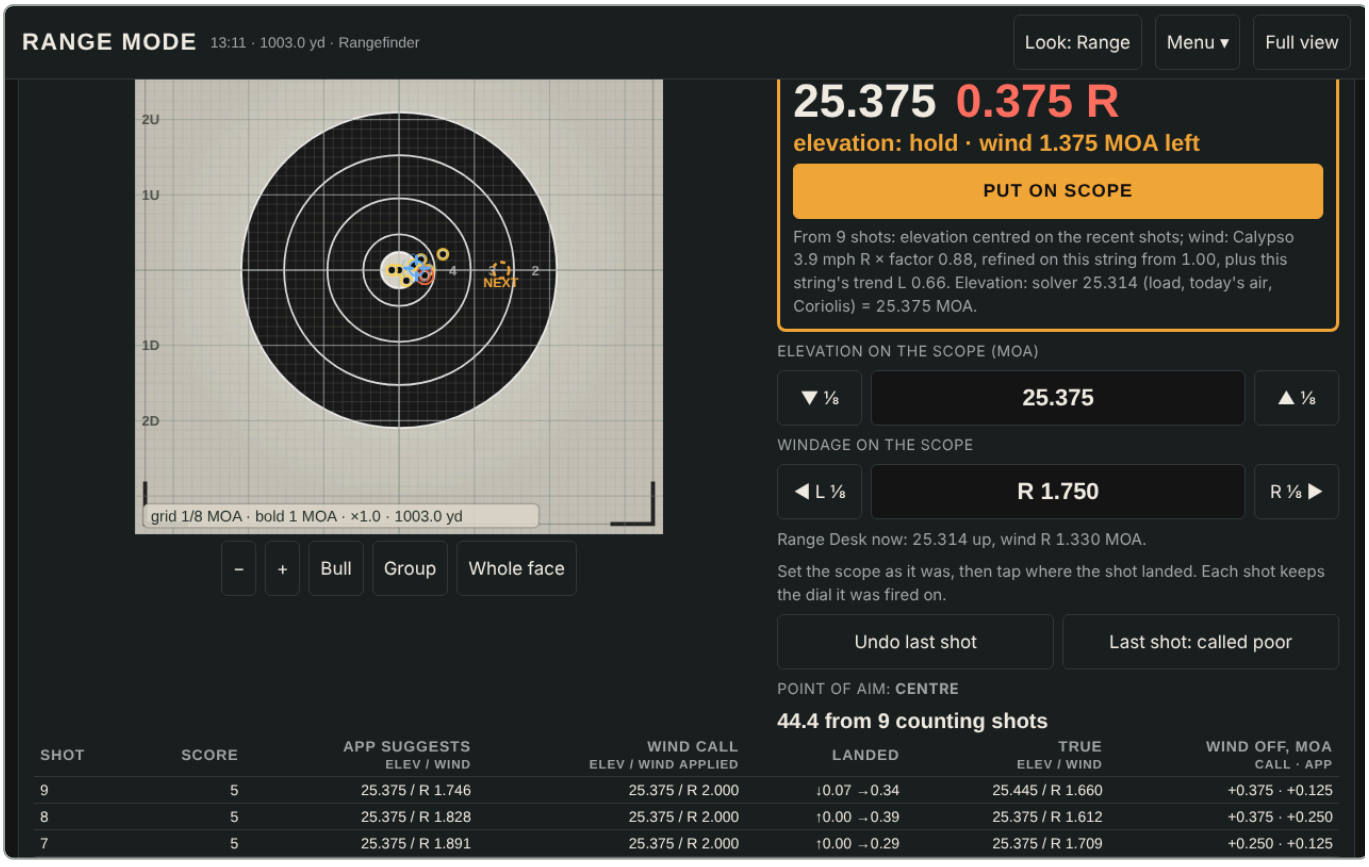


Before a string: the checklist, with Set or Change opening one card.

During the string

Only the shoot is shown: the target, **Next shot**, the dial steppers (each step is your scope's ⅛ or ¼ MOA) and the last three shots. **More** opens the set-up cards, for example to change the mirage.

1. Set the scope as fired, then **tap where the shot landed**, from ShotMarker's screen, your plot sheet or the spotting scope. Each shot keeps the dial it was fired on.
2. **Next shot** gives the elevation and windage to put on, and the change in MOA. **Put on scope** sets it in one tap. The reason line explains it, including the elevation's parts (solver, truing and string trend).
3. On the face: the newest shot ripples; a **reticle** marks the group centre, with a circle at its average spread; the dashed **NEXT** ring shows where the next shot should land on the current dial.
4. **Last shot: called poor:** the shot still scores but is left out of all truing, and shows ×.
5. **Point of aim:** when aiming off, step it in ½ MOA before the shot. An AIM mark shows, and shots are judged from it until you choose Centre.
6. **Undo last shot** removes a mistaken tap.
7. **End string** after the last counting shot. It logs everything in one push, ticks the detail off, saves the ranged distance, and offers the next detail.



During a string: target, Next shot, dial and the per-shot table.

Table column	Meaning
App suggests	What Next shot advised for that shot
Wind call	What was on the scope: the shooter's or coach's call
Landed	Where the shot went, from the point of aim
True	The elevation and wind the shot actually needed
Wind off · call · app	How far the call and the app were each from True, in MOA

Entering a detail afterwards, from your plot sheet

Use this when you can't use the tablet during the detail, for example under competition rules.

1. Right after the detail, open **Entering a detail shot earlier?** above Start string, and enter the time the detail **started** and **finished**.
2. Tap **Start string**, and set the sighters and the number to count, for example 2 + 15.
3. Work down your sheet row by row (A, B, then 1, 2, 3 and so on). For each row, set the **elevation (EVN)** and **windage (Wind call)** with the steppers to what was on the scope, then **tap where that shot is plotted**. Use Bull or Group to zoom, and the MOA scales to place it. Mark **Called poor** or set **Point of aim** as on the sheet.
4. Tap **End string**. The string is dated to the times you entered. It takes the forecast for that hour and, if this device was connected to the Calypso during the detail, the **Calypso average for exactly those minutes**, which the app records through the day. Today's wind factor is then learned as if you had entered the detail live.

Leave the tablet or phone connected to the Calypso, placed near your firing point, through the shoot. Enter each detail as soon as you can afterwards.

After the string

The **wind bracket** shows the lightest and strongest wind met, the windage for each, and the single setting that would have dropped the fewest points.

Today's wind factor, carried into the next detail. At End string, the app compares the crosswind your counting shots actually met with what the forecast, the Calypso and your Kestrel said over the same time. For example, a detail that met 3.6 mph left against a forecast of 6.3 mph left gives the forecast a factor of 0.74. The factor is carried **pro rata**: the next detail's opening windage is its own forecast × today's factor, worked out at its own distance and time. So the lesson of one detail reaches the next, not the MOA value itself.

- You see it on the End string banner ("Today's wind factor: forecast ×0.74 …") and on the **Next detail** card, whose opening already includes it.
- Next shot starts from the same trued opening, and the Match plan's later details update too, so a reprinted sheet carries it.
- Each detail refines it, weighted by counting shots and drawn towards 1, so one odd detail moves it only part of the way. Called poor shots are left out.
- It is kept per source and per day, and starts afresh each morning. Untick **Use today's wind factor** on the Next detail card if the conditions change completely, for example when a front comes through.

8 TRUING: VELOCITY, BC AND WIND

- **Correct from impact** (Range mode card): the group needed more or less elevation than was dialled. The app attributes it to **velocity** below 800 yd and to **drag (BC)** from 800 yd, holding velocity at the Garmin reading. Use **Up** or **Down** in MOA, or **Use the plotted group**, then **Add correction**. **Save to load** makes it permanent, and **Undo last save** reverses it.
- **BC from velocity** (Records): the Garmin muzzle velocity plus ShotMarker's speed at the target give the BC directly, string by string. On 31 October: G7 0.366 against 0.349. **Use in load** applies it (with Undo); **Copy for QuickLOAD** gives a summary to paste.
- **Wind truing** (Range mode card): compares each wind source (Calypso, model, consensus, Kestrel) with the wind your bullets met, and fits a factor per firing point. Tap **Use** to apply it to that source.
- **Today's wind factor** trues from one detail to the next across the day (section 7).
- **Next shot** trues during a string from your recent shots, weighted towards the latest, and refines the Calypso factor on the string itself.

9 INSTRUMENTS

Instrument	How
Calypso wind meter (WDOT)	Range mode → Wind → connect over Bluetooth (Chrome on Android). Close Anemotracker first. The speed units are being confirmed against the Kestrel; choose the scale that matches.
Weatherdot	Range mode → Atmosphere → **Connect Weatherdot (live)**, for temperature, pressure and humidity.
Kestrel	Enter its readings under Atmosphere (Kestrel / Weatherdot) and Wind (Kestrel / mine), or use **Read Kestrel photo**.
Garmin chronograph	**Read Garmin photo** after a string, or import the ShotView export later.
Sig rangefinder	Type the line-of-sight distance, or use **Read rangefinder photo**.

10 CAPTURING FILES FROM SHOTMARKER

ShotMarker keeps each string on its hub for about a year, but copy it at the range the same day. The CSV holds each shot's exact position and speed; the .tar is a full copy.

1. **Bring a laptop or an Android phone** that lets you save and share files.
2. Join the **ShotMarker Wi-Fi** for the hub (scan the QR code on the unit), open a browser and go to **192.168.100.1**.
3. Open the **String Archive** from the menu. Find the day's strings by date and time; note that the ShotMarker clock may run a few minutes off.
4. **Tick your strings** and tap **Export CSV**. If it only exports the string on screen, open each one and export it.
5. With the same strings ticked, tap **Export**. This saves the .tar.
6. **Screenshot** each string's target screen, and the face settings if in doubt.
7. **Check Downloads** before leaving: a .csv (for example `SM_shotlog_Nov_7.csv`) and a .tar. Open the CSV: one row per shot with time, x, y, score and speed.
8. **Don't delete, re-save, edit or import** anything on the hub. Export when the range isn't shooting, as exporting can slow the system.

Also export the **Garmin ShotView** sessions (.xls, .xlsx or .csv) from the phone app.

11 IMPORTING RESULTS

1. Full view → **Plotted shots** → **Import results**.
2. Choose the **ShotMarker .tar** and the **Garmin file** together. Strings are matched by the timing between shots, so device clocks and names don't need to agree.
3. Check the **dial** for each string (elevation and windage as fired) and tap **Import**.
4. Shots appear on the face, and each string gets its velocity, speed at the target and BC from velocity. Where you had already entered the string (live or from your plot sheet), ShotMarker's exact positions **replace your tapped shots one for one**, keeping the dial, called shots and aim you entered, so nothing is counted twice. The app allows for the ShotMarker clock being a few minutes out. A ShotMarker CSV goes through **Import a ShotMarker CSV**, below it.

12 RECORDS, BC AND DATA FILES

- **Records** (full view header, or Range mode → Menu): every string, filtered by date, shooter, session, event, distance, load, coach or batch.
  - **Summary:** strings, shots, points and V count, average group, average Garmin velocity, and the elevation needed against the app.
  - **By batch:** each batch worked out on its own. If untagged strings look like two batches, **Tag as two batches** is offered.
  - **Tap a string** to edit its tags or show it on the target.
  - **Export** the strings or their shots as CSV, **save them as a data file**, or delete them from the device once saved. A storage meter shows how full the device is.
- **Data files** (Plotted shots → Data files): **Save all data to a file** after every session, and use **Open a data file** to restore or merge from another device. Duplicates are skipped.
- **Show** (above the face, with ◄ ►) steps through strings one at a time, each on its own face at its own distance.

13 FULL VIEW REFERENCE

Every card folds to its title; tap the title to open or close it, and the app remembers the choice.

Card	Use
Map and header	Search a place; **Place firing point** and **Place target** set today's pins; satellite imagery
Match plan	Ranges, Rifles, squadding details, forecasts, print sheets (sections 3–5)
Load	Choose, edit, add, duplicate or delete loads (section 2)
Distance truing	Nominal card distance, map distance and rangefinder distance; the true one is used
Ballistic solution	Elevation and windage, with a table for all distances; options for spin drift, aerodynamic jump and Coriolis. **Highest point of flight** gives how far above the line of sight the bullet rises and where, for example 88 in at 537 yd for 1000 yd (hold a finger on it for centimetres).
Line of fire	Bearing, angle and the wind relative to your line
Wind on the line / Atmosphere	Choose the source: Kestrel or mine, Calypso, model, consensus or airfield
Model comparison, Airfield, Sun and light	Background for the wind and light picture. **Open Windy for this range** (in Wind on the line) opens Windy's forecast animation in a new tab.
Target face	Face system and rings (ICFRA and Bisley), true-size shots
Plotted shots	Shots, validation table, imports, data files, CSV

14 NO SIGNAL, BACKUPS AND TROUBLESHOOTING

- **No signal:** the app, the last weather (marked "saved, no signal") and the saved map work offline. The **Offline** chip shows when the app is offline.
- **Backups:** save a data file after each session, and keep it off the tablet as well.
- **New buttons missing after an update?** GitHub may still be serving the old file. Wait a few minutes and open the address with `?v=` and a new number.
- **A Bluetooth instrument won't connect?** Close its own app first, check it's charged and nearby, and use Chrome on Android.
- **A shot was tapped in the wrong place?** Use **Undo last shot**, or mark it **called poor** if it was a bad shot.
- **Competition rules** restrict electronic aids at the firing point. You choose when to use the app; it records in the background either way.